Skeletal Traction for Femur Shaft Fractures

Ortho Team

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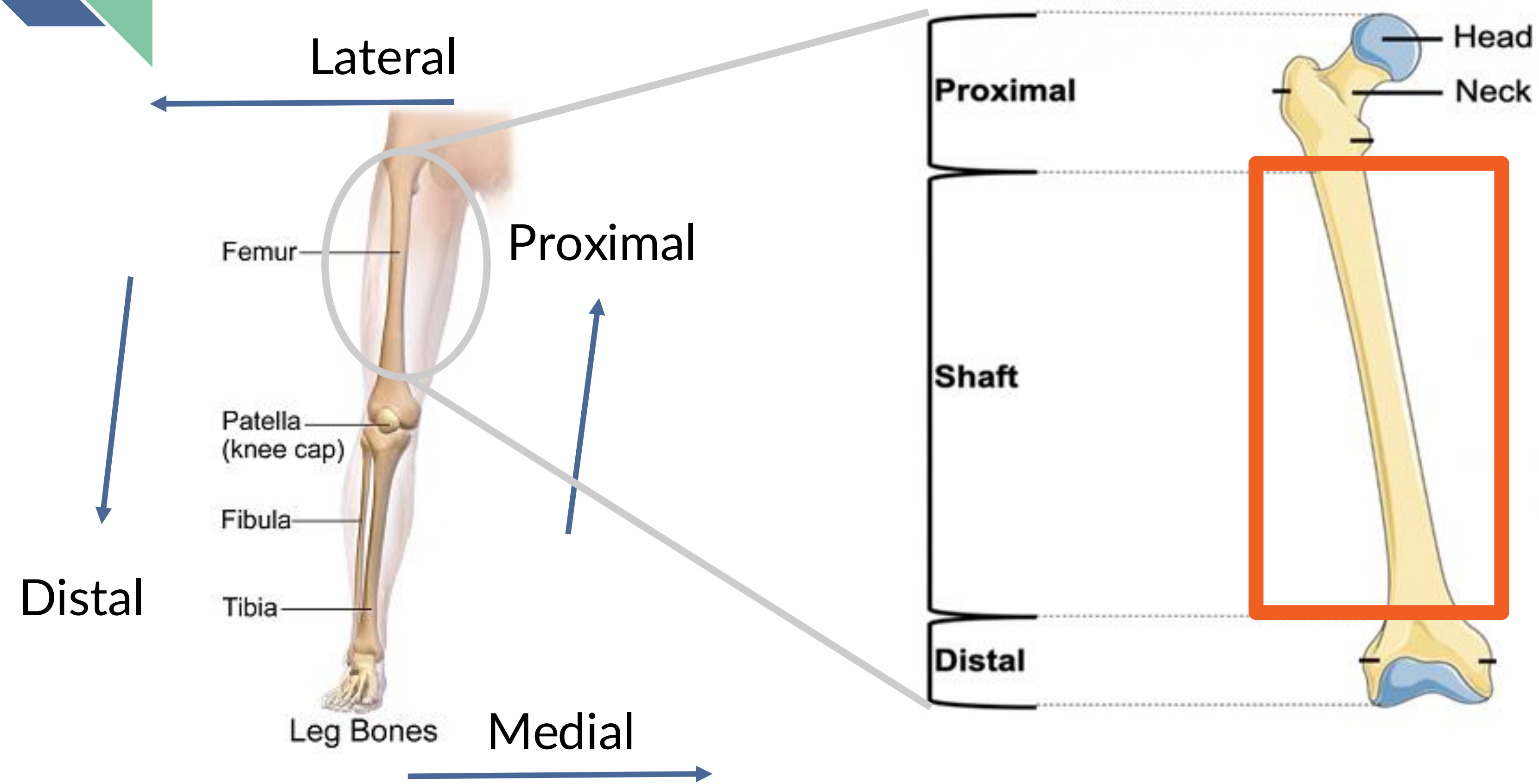
Outline

1. Background and Problem
2. Need Statement
3. Stakeholder Analysis and Value Proposition
4. Design Strategy
5. IP search and Initial Prototypes

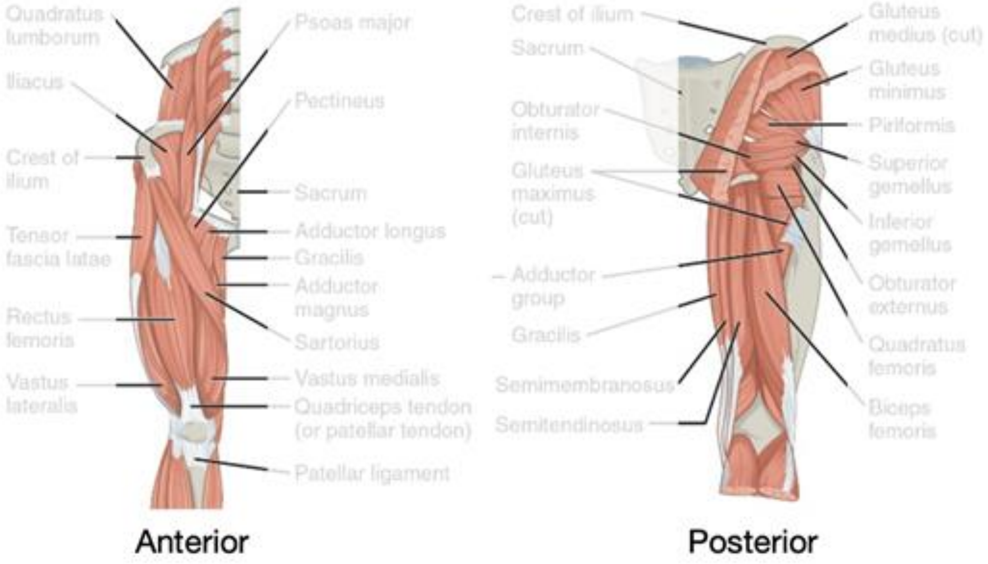
Background and Problem

Anatomy

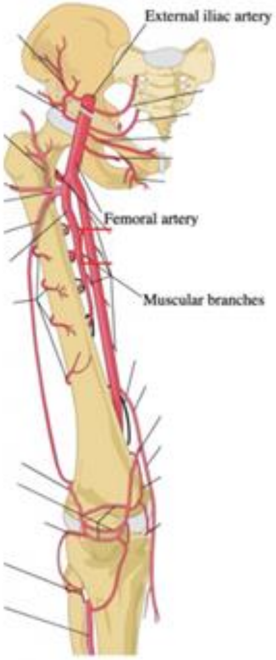


Lateral
Femur
Patella
(knee cap)
Fibula
Tibia
Leg Bones
Proximal
Distal
Medial
Proximal
Shaft
Distal
Head
Neck

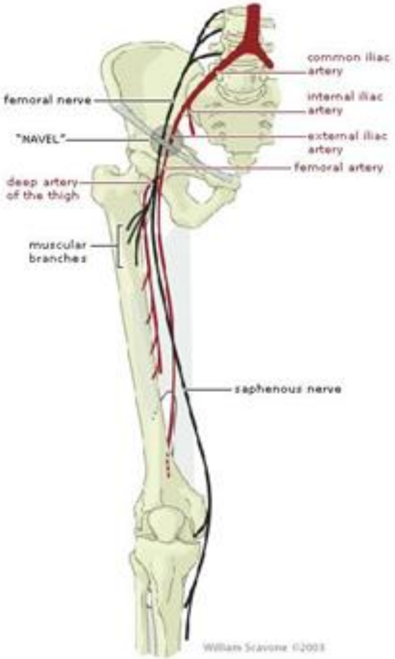
Anatomy



Quadratus lumborum
Psoas major
Iliacus
Pectineus
Crest of ilium
Sacrum
Tensor fascia latae
Adductor longus
Gracilis
Adductor magnus
Rectus femoris
Sartorius
Vastus lateralis
Vastus medialis
Quadriceps tendon (or patellar tendon)
Patellar ligament
Anterior
Crest of ilium
Sacrum
Obturator internis
Gluteus maximus (cut)
Adductor group
Gracilis
Semimembranosus
Semitendinosus
Gluteus medius (cut)
Gluteus minimus
Piriformis
Superior gemellus
Inferior gemellus
Obturator externus
Quadratus femoris
Biceps femoris
Posterior



External iliac artery
Femoral artery
Muscular branches

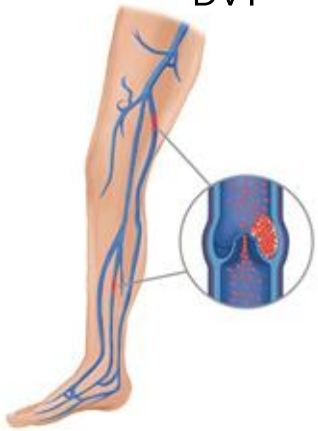


common iliac artery
femoral nerve
internal iliac artery
"NAVEL"
external iliac artery
femoral artery
deep artery of the thigh
muscular branches
saphenous nerve

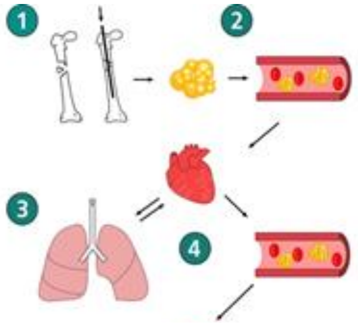
Complications

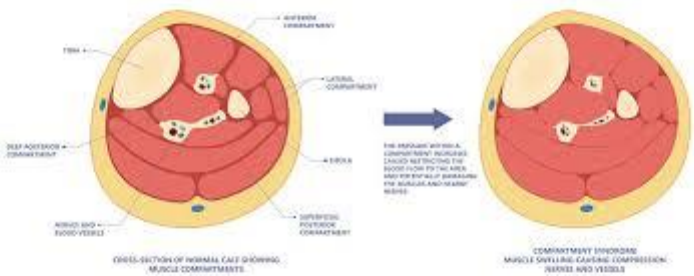
Hemorrhage

DVT

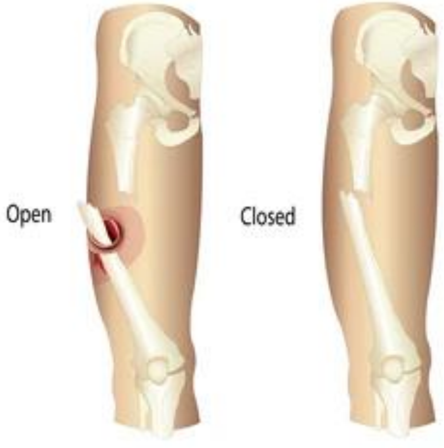


Fat Embolism



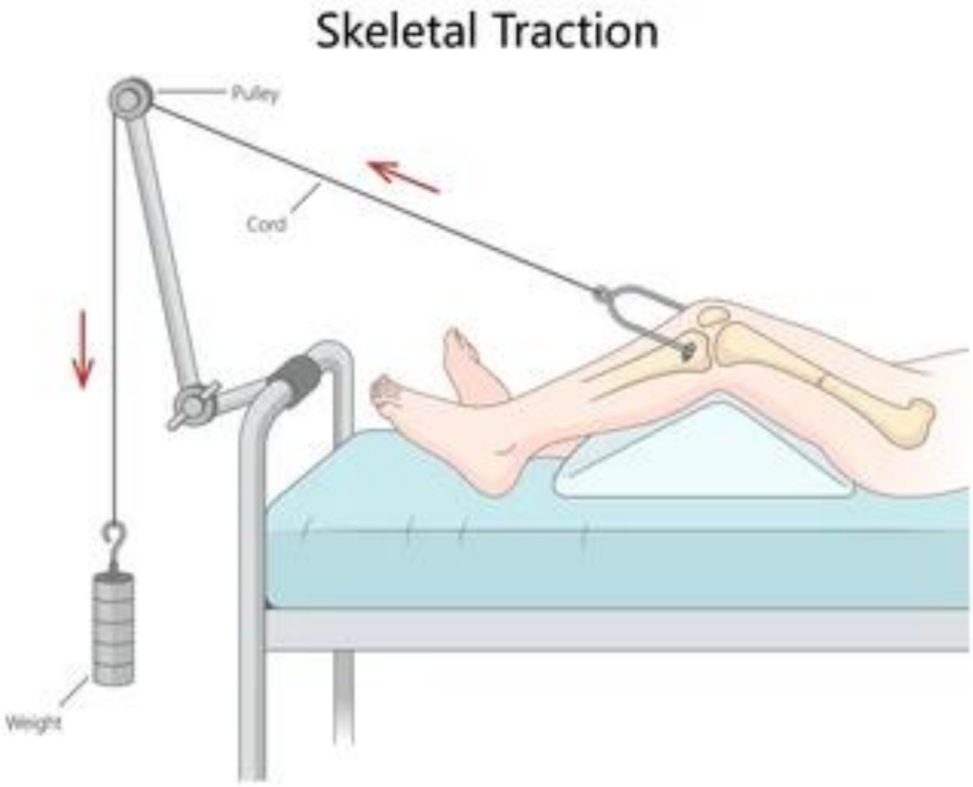


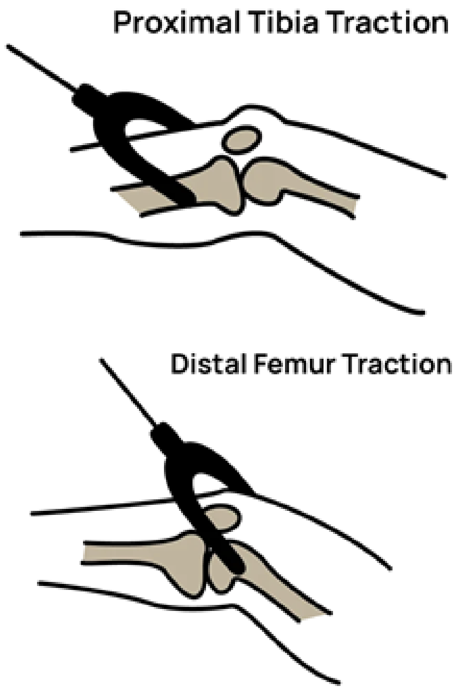
Compartment Syndrome



Further damage

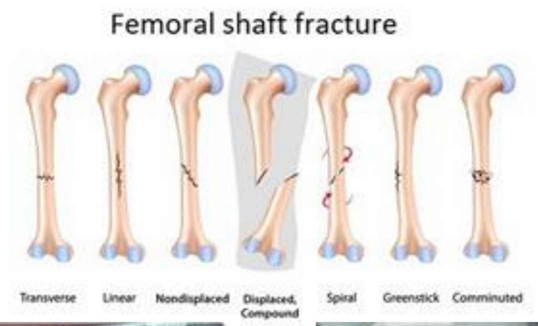
Reducing Complications





Standard of Care

1. Examination
2. X-Rays to confirm diagnosis (CT if necessary)
3. Order traction
    a. Bed, traction frames, weights
4. Apply traction
5. Additional scans
    a. Traction weight removed
    b. Special traction bed
    c. Makeshift traction apparatus
6. Surgery for internal fixation
    a. Inaccurate scans
    b. Length and stability not maintained - harder, longer realignment; administer more drugs



Causes

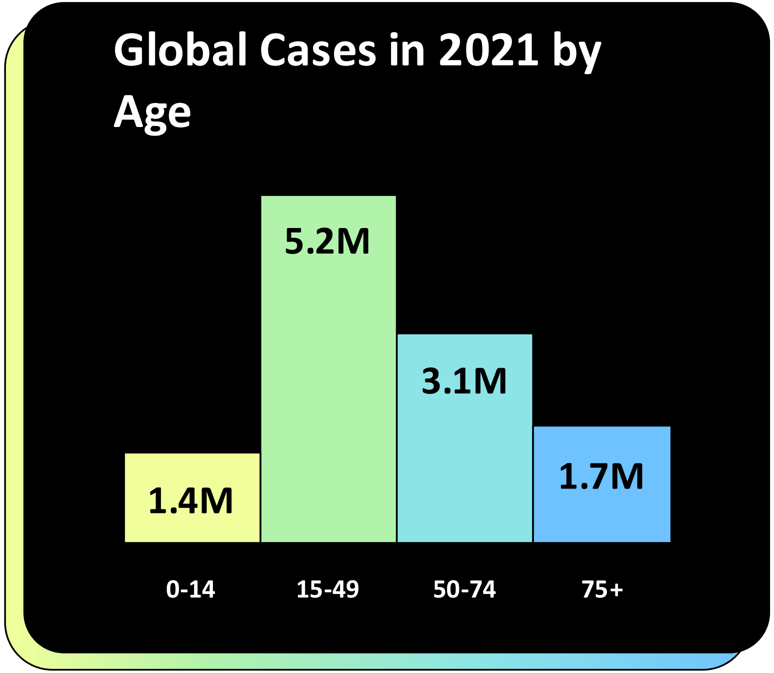
High Impact

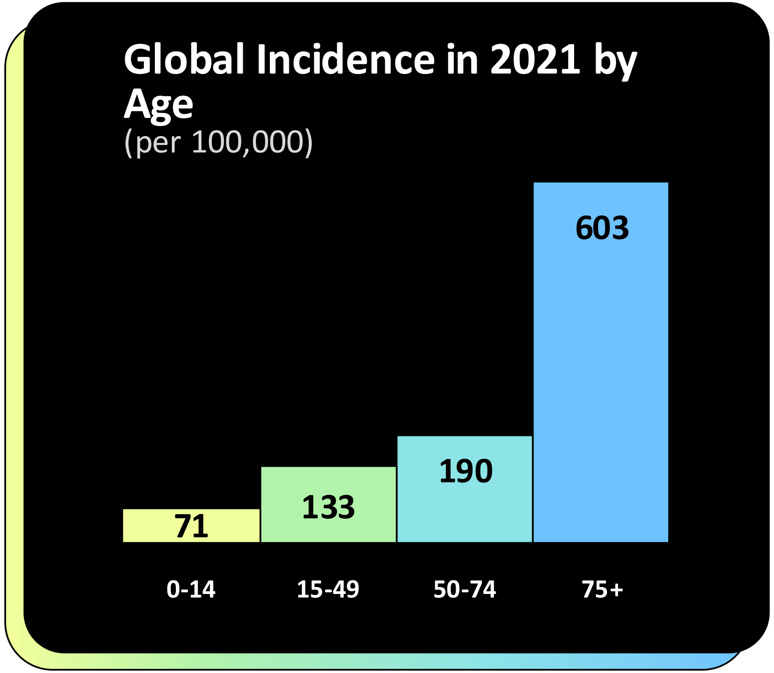
Low Impact

Femur shaft fractures by the numbers

Global cases: 11.5M

Global incidence: 146 (per 100,000)





Fu F, Liu B, Pu H, Wang Y, Zhang P, Wei S, Gu H, Zhang Q, Ye H. Global Trends in the Incidence and Primary Causes of Femoral Fractures, Excluding Femoral Neck Fractures: A Global Epidemiological Study. *Risk Manag Healthc Policy*. 2025;18:117-129 https://doi.org/10.2147/RMHP.S498918

Market Size

ARBUTUS MEDICAL

$615 SAVINGS PER CASE

STANDARD WORKFLOW $1,744	COST TABLE	WORKFLOW WITH TrakPak® $1,129
$420	Tray Sterilization[1,5] Power System • Battery Pack • Traction Pin • Kirschner Bow	$829 Cost of TrakPak®
$54	Capital Depreciation & Maintenance[2,3,4,5]	
$70	Single Use Materials[5] Prep Materials • Dressing Supplies	
$900	Staff Time Gathering & Opening[5,6,7,8,9]	$100
$300	Staff Time Applying Traction[5]	$200

US Population in 2021

330 Million

Hip and Femur Fracture Cases

690,000

Femoral Shaft Fractures Treated with Traction

~23%*

* varies widely by location

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Market Size (conservative)

Total Addressable Market (TAM)

690,000 cases x 15% = 103,500

Revenue/sale	**$600**
Sales/year	**103,500**
Total	**$62.1M**

Serviceable Available Market (SAM)

390,000 cases x 15% = 58,500

Revenue/sale	**$600**
Sales/year	**58,500**
Total	**$35.1M**

Need Statement

Need Statement

Statement

A way to provide measurable/quantifiable skeletal traction for femur fractures that can be easily moved with the patient so constant traction can be applied and easily managed throughout treatment/to inform further treatment

Feedback

- Too general
- Focus (scope down)
- Identify the type of fracture
- Measurable outcome

Need Statement

Statement

A way to increase time in constant skeletal traction, from the time of initial diagnosis to the time of surgery, for patients with femoral shaft fractures in order to maintain the proper length between the hip joint and knee joint.

Feedback

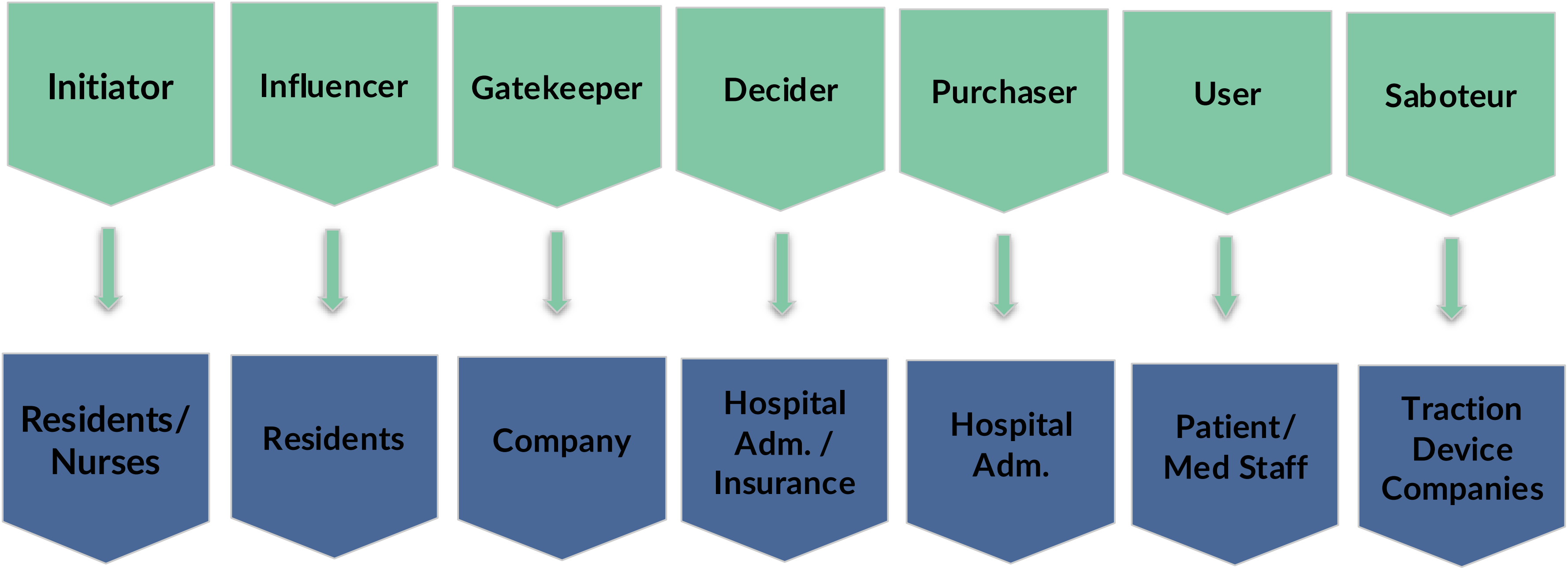
- Narrative
- Why traction is taken off.
- Increased time ?
- Proper length

Need Statement

- A way to maintain stable tension during skeletal traction that is robust to patient position, from the time of initial diagnosis, to scans, to the time of surgery, for patients with femoral shaft fractures to prevent femur displacement that can cause further damage.

Stakeholder Analysis and Value Proposition

Stakeholder Analysis



Empathy Maps

HEAR 👂

- Painful
- Adjusted multiple times

1

THINK AND FEEL 🧠

- Effectiveness of treatment
- Painful?
- Recovery time

2

SEE 👀

- Skin traction solutions
- Adjustment needed for skeletal traction

3

SAY AND DO 🗣

- Hard to maneuver
- Outcomes vary

4

Value Proposition

Jobs

- Movable/ Portable traction
- Stable traction

Pains

- Adjusting equipment
- Removing for X-Ray
- Not providing the correct treatment

Gains

- Tractions moves with patient
- Consistent measurable traction
- More efficient Surgery

Value Proposition

Product/Services

A traction system that is constant and can be moved with the patient at all times

Pain Relievers

- Not adjusting the traction system
- Device not removed for X-Ray

Gain Creators

- Effective treatment (outcome)
- Traction can be adjusted by nurses
- Better performance

Design Strategy

Metrics of Success

Primary Metrics

1. Measurable/Adjustable Traction

2. Self Contained Traction

3. Portable

Secondary Metrics

4. Low Cost

5. Easy Application

Metrics of Success

1. **Measurable and Adjustable Traction**
   - Tension measurement is accurate within 0.5 lbs of actual
   - Amount of traction can be adjusted without removal

1. **Self Contained Traction**
   - Maintain any tension between 5-40 lbs without weights

1. **Portable**
   - Can be moved and stays with patient, does not need to be removed

Metrics of Success

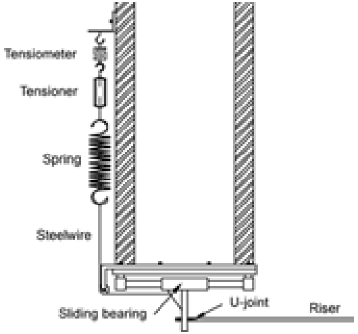
4. **Low Cost**
    - <$400 to produce

5. **Easy Application**
    - Can be applied by nurse or resident in 5-10 min

Potential Approach

Measurable and Adjustable Traction

- Load cell/pins, amplifier, voltage convertor
- Microcontroller, display, LEDs, buzzer
- Dial or crank tightens/loosens rope or springs



Self Contained Traction

- Pulley or spring system to apply tension
- Connected to brace at the thigh and pin at the knee

Portable Traction

- Components conform around patient, not attached to bed

Feasibility and Barriers to Adoption

Feasibility

- Incorporates elements from existing systems
- Does not require additional infrastructure

Barriers to Adoption

- Higher upfront costs
- Insurance reimbursement
- Initial learning curve
- Lack of reliability data

Resources Needed and Constraints

Specialized Resources

- No need for special equipment, complex parts, detailed tissue models anticipated
- Leg models can be simulated with physical approximations

Expertise

- Extensive research into biomechanics and safely applying force to leg required
- Might need to consult trauma surgeons or residents

IP search and Initial Prototypes

Brief Regulatory Overview (FDA)

Solution Regulatory Overview

- Invasive Skeletal traction device
- FDA Identification: § 888.3030
- FDA Product Codes
  - Traction Device Fixtures: KTT
  - Traction Invasive pins: JEC

Risk Classification

- Falls under class 2 → 510(K) Pathway
- Must prove substantial equivalence to predicate device

§ 888.3030

(Single/multiple component metallic bone fixation appliances and accessories)

Brief IP search

Freedom to operate

- Thomas Traction system → Invented during WW1 (Expired)
- Components: Steinmann pins, Weights, static frames

Spring Traction

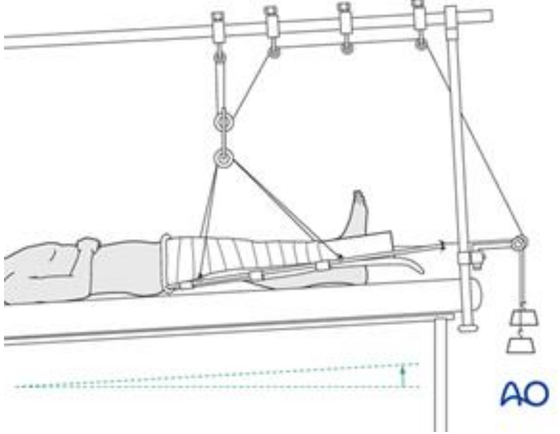
- Device for pelvic traction: US4865022A
- Self applicable spring loaded pelvic traction device → Patent expired

Elastic Band Traction

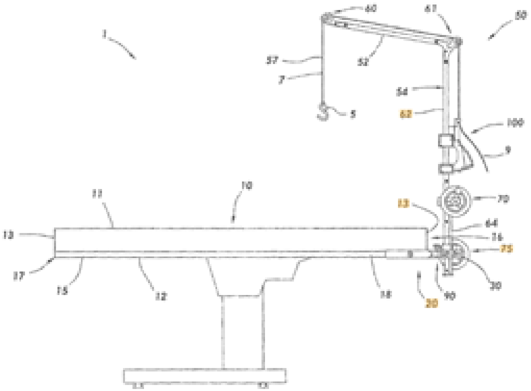
- Patents exists: Elastic band traction of spine
- FTO: Elastic band specifically for femoral traction

Recent Innovation

- Surgical traction tension adjustment device: US11813188B2
- Active Patent on winch like traction device that tensions cable



Thomas splint system is open domain



Winch-like traction device that attaches to a bed rail and tensions cable to apply traction

Predicate devices

QuikBow Traction kit

- Rapid and sterile skeletal traction device
- Traction kit is disposable (expensive)

Traction beds + Accessories

- Components (sold individually)
    - Beds, frames clamp bars, clamps, bars, pins, pulleys, and weights
- Companies
    - Stryker, Hill Rom, Arthrex

Summary

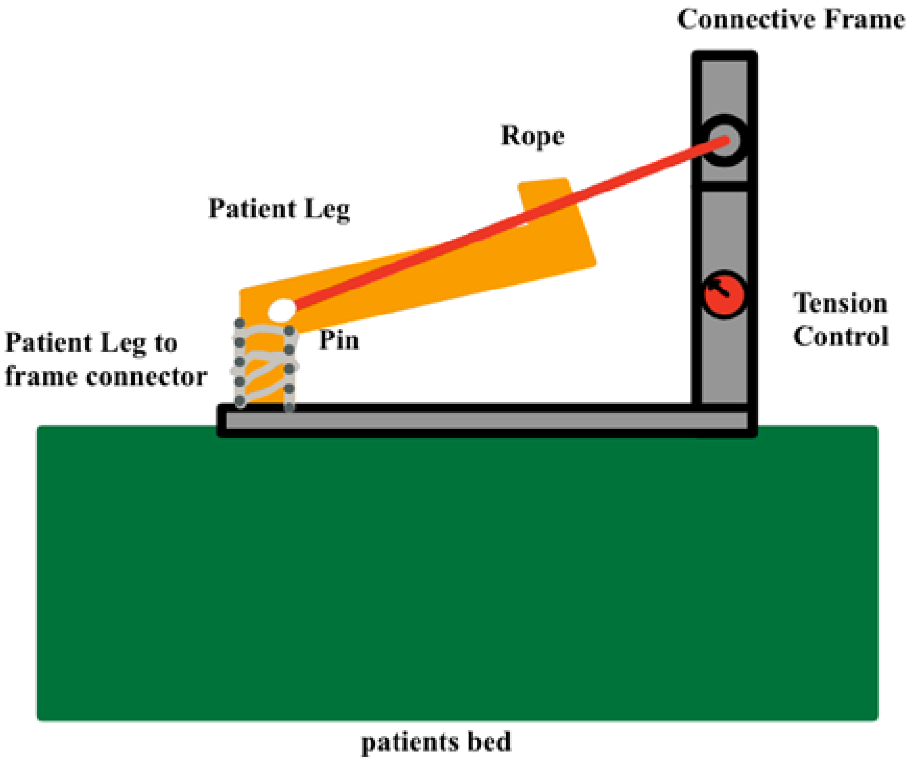
- Predicates: Single use bow technology on market
- Reusable: little deviation from standard thomas splint

QuikBow Traction kit (spring based) for quick portable skeletal traction (single use)

Prototype Concept: Spring Tensiometer

Spring Tension Solution

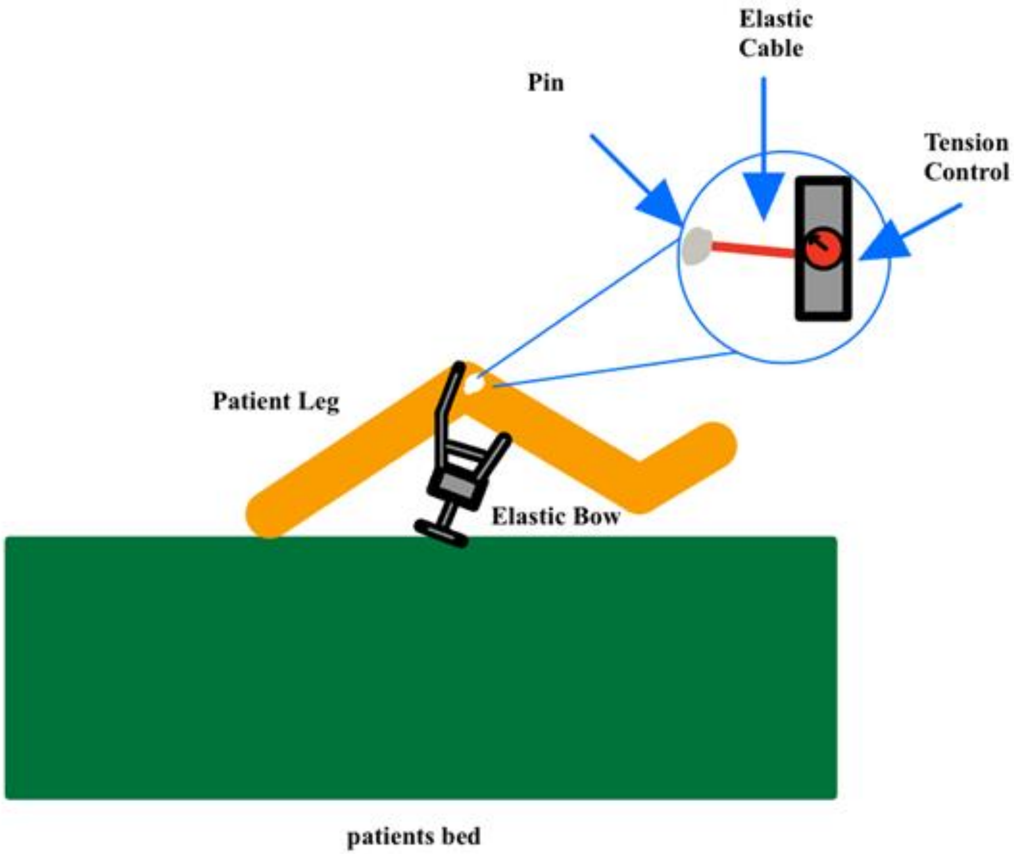
- Concept
  - Spring system applies controlled tension to rope
- Components
  - Tensiometer
  - Tensioner
  - Spring
  - Moveable Connective Frame
- Considerations
  - Spring Creep and Strength



Prototype Concept: Elastic Tension Cables

Elastic Tension Cables Solution

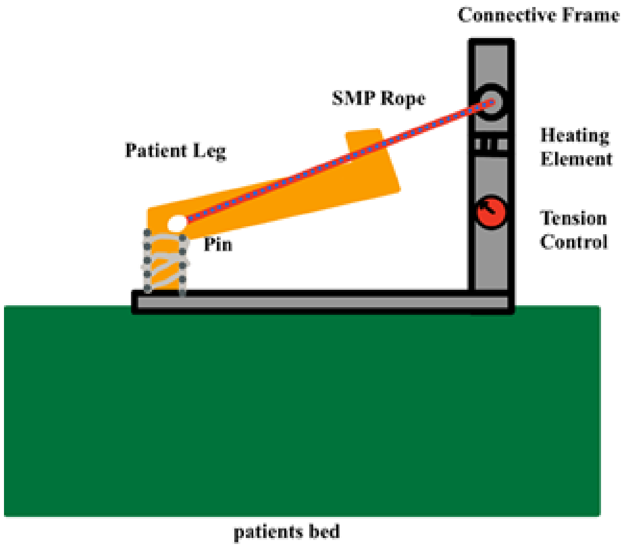
- Concept
  - Bow technology
  - Elastic cable applies tension to leg
  - Dial used to adjust tension in cable
- Components
  - Elastic Cable
  - Connective Frame
  - Tension Control Dial
- Considerations
  - Elastic Cable Creep
  - Measuring cable tension (method)
  - Reusable Bow

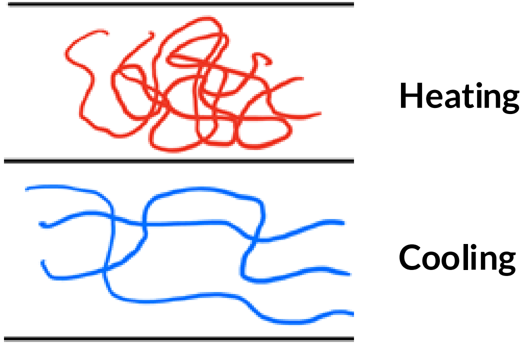


Prototype Concept: Shape Memory Polymers

Shape Memory Polymers

- Concept
  - Shape memory polymer (SMP) rope
  - Heat to reduce tension
- Components
  - SMP Rope
  - Heating component
  - Tension Control
- Considerations
  - Material strength
  - Heating in OR
  - Material transition temperature





Questions to be answered

Market Research

Profitability

- Would hospitals buy it
- Margins for economical competitivity

Specific Design

Select Mechanism

- Tension cables, Spring, SMP, etc
- Balancing portability & traction capabilities

Regulatory Understanding

IP Research

- Further research IP landscape
- Develop product in FTO space

References

[1] Orthopedic traction bed. Arcan Medical Co., Ltd. (2025, November 29). https://www.arcanmedical.com/product/orthopedic-traction-bed/#:~:text=%E6%8F%8F%E8%BF%B0\

[2] Request access. Federal Register :: Request Access. (n.d.). https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-888/subpart-D/section-888.3030

[3] Google. (n.d.). US4865022A - self applicable spring loaded pelvic traction device. Google Patents. https://patents.google.com/patent/US4865022A/en

[4] Twittercreator. (n.d.). Peter V Giannoudis. for Simple, transverse, middle 1/3 fractures. https://surgeryreference.aofoundation.org/orthopedic-trauma/adult-trauma/femoral-shaft/simple-transverse-middle-1-3-fractures/temporary-thomas-splint

[5] Google. (n.d.-b). US8905952B2 - simple portable lumbar spine distraction device and method. Google Patents. https://patents.google.com/patent/US8905952B2/en

[6] Google. (n.d.-c). US11813188B2 - surgical traction tension adjustment device. Google Patents. https://patents.google.com/patent/US11813188B2/en

[7] Traction accessories. Kinetec USA. (n.d.). https://www.kinetecusa.com/product/orthopaedic-traction/traction-accessories/

References

https://study.com/academy/lesson/how-to-apply-a-traction-splint.html
https://www.istockphoto.com/search/2/image?mediatype=illustration&phrase=motorcycle+accident+cartoon
https://www.istockphoto.com/search/2/image?mediatype=illustration&phrase=old+man+falling&tracked_gsrp_landing=https%3A%2F%2Fwww.istockphoto.com%2Fillustrations%2Fold-man-falling
https://www.emdocs.net/an-overview-of-femur-fractures-location-imaging-complications-and-management/
https://anatomy.host.dartmouth.edu/HAE/Lowerextremity/tutorials/nerves/femoral.html
https://www.physio-pedia.com/Femoral_Shaft_Fractures
https://link.springer.com/article/10.1186/s13018-019-1145-6
https://www.anatomyatlases.org/firstaid/ThighInjury.shtml
https://www.shutterstock.com/search/open-fracture?image_type=vector&dd_referrer=https%3A%2F%2Fwww.google.com%2F
https://arbutusmedical.com/blog-why-or-why-not-skeletal-traction/
https://arbutusmedical.com/inefficiencies-of-skeletal-traction-in-the-emergency-department/#:~:text=Streamline%20the%20Procedure,which%20provides%20many%20benefits%20including:.&text=Considering%20the%20cost%20of%20TrakPak,traditional%20methods%20(Figure%208).
https://www.dovepress.com/global-trends-in-the-incidence-and-primary-causes-of-femoral-fractures-peer-reviewed-fulltext-article-RMHP#t0001
https://www.ncbi.nlm.nih.gov/books/NBK556057/?utm_source=chatgpt.com
https://www.census.gov/newsroom/press-releases/2021/2021-population-estimates.html